



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2

Division A

Subject: Programming with Java

ASSIGNMENT No : 13

TITLE : Write a Java program to perform reading from and writing to a file using file handling operations.

Problem Statement

Write a Java program to perform reading from and writing to a file using file handling operations.

Learning Objectives

To perform file reading and writing operations using Java file handling classes.

Theory

Java provides file handling through the **java.io** package. Classes such as **File**, **FileReader**, **FileWriter**, **BufferedReader**, and **BufferedWriter** are used to perform input and output operations. File handling enables permanent storage and retrieval of application data. Proper exception handling is necessary to manage file-related errors. These concepts are essential for data persistence.

Outcome

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp13/Main.java
Hello, this is a Java file handling program.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

CODE

```
1 package Exp13;
2
3 import java.io.*;
4
5 public class Main {
6     public static void main(String[] args) {
7         try {
8             FileWriter fw = new FileWriter("Exp13/data.txt");
9             fw.write("Hello, this is a Java file handling program.");
10            fw.close();
11
12            FileReader fr = new FileReader("Exp13/data.txt");
13            BufferedReader br = new BufferedReader(fr);
14
15            String line;
16
17            while ((line = br.readLine()) != null) {
18                System.out.println(line);
19            }
20
21            br.close();
22            fr.close();
23
24        } catch (IOException e) {
25            System.out.println("File error: " + e.getMessage());
26        }
27    }
28 }
```

EXERCISE 1

```
1 package Exp13.Exercise;
2
3 import java.io.*;
4
5 public class Exp1 {
6     public static void main (String[] args) {
7         try {
8             FileWriter fw = new
FileWriter("Exp13/Exercise/employee.txt");
9
10            fw.write("Employee ID: 101\n");
11            fw.write("Name: Rahul\n");
12            fw.write("Salary: 50000\n");
13
14            fw.close();
15
16            FileReader fr = new
FileReader("Exp13/Exercise/employee.txt");
17            BufferedReader br = new BufferedReader(fr);
18
19            String line;
20
21            while ((line = br.readLine()) != null) {
22                System.out.println(line);
23            }
24
25            br.close();
26            fr.close();
27
28        } catch (IOException e) {
29            System.out.println("File error: " + e.getMessage());
30        }
31    }
32 }
```

EXERCISE 2

```
1 package Exp13.Exercise;
2
3 import java.io.*;
4
5 public class Exp2 {
6     public static void main (String[] args) {
7         try {
8             FileWriter fw = new
FileWriter("Exp13/Exercise/patient.txt");
9
10             fw.write("Patient ID: 101\n");
11             fw.write("Name: Amit\n");
12             fw.write("Age: 25\n");
13             fw.write("Diagnosis: Fever\n");
14
15             fw.close();
16
17             FileReader fr = new
FileReader("Exp13/Exercise/patient.txt");
18             BufferedReader br = new BufferedReader(fr);
19
20             String line;
21
22             while ((line = br.readLine()) != null) {
23                 System.out.println(line);
24             }
25
26             br.close();
27             fr.close();
28
29         } catch (IOException e) {
30             System.out.println("File error: " + e.getMessage());
31         }
32     }
33 }
```

Exercise

1. Create a program to write and read **employee details from a file**.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp13/Exercise/Exp1.java
Employee ID: 101
Name: Rahul
Salary: 50000
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ █
```

2. Create a Java program to write **patient details** (patient ID, name, age, and diagnosis) to a file and read the details from the file for display.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp13/Exercise/Exp2.java
Patient ID: 101
Name: Amit
Age: 25
Diagnosis: Fever
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ █
```

Conclusion

This experiment helped in understanding the concept of file handling in Java using the `java.io` package. It demonstrated how to perform reading and writing operations using classes such as `FileReader`, `FileWriter`, `BufferedReader`, and `BufferedWriter`. The experiment also showed how file handling can be used for permanent storage and retrieval of data. Overall, it improved understanding of data persistence and the importance of handling file-related errors properly.